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Article

Heparinized Dendritic Mesoporous Silica Nanoparticles for the Selective Adsorption and Isolation of Thrombin

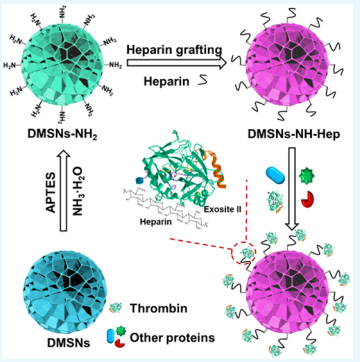
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ABSTRACT: A novel solid-phase extractant, heparinized dendritic mesoporous silica nanoparticles (DMSNs-NH-Hep), is prepared by grafting heparin onto amino-terminated dendritic mesoporous silica nanoparticles. DMSNs-NH-Hep exhibit a spherical morphology and possesses a unique central-radial pore structure with a BET surface area and pore volume of 230.81 $m^2 g^{-1}$ and 1.23 $cm^3 g^{-1}$, respectively. Due to the specific interaction between heparin and thrombin, DMSNs-NH-Hep demonstrate favorable adsorption selectivity and a rapid adsorption rate toward thrombin. In PBS buffer at pH 7.4, an adsorption efficiency of 81.1% for 0.1 mL of thrombin solution (85 U/mL) is obtained by using 0.06 mg of DMSNs-NH-Hep in 3 min. An adsorption capacity of 237.35 U mg^{-1} is derived for thrombin. The captured thrombin can be readily eluted by a 0.3% sodium dodecyl sulfate (SDS) solution with an elution efficiency of 80.4%. The potential of DMSNs-NH-Hep in complex sample analysis is confirmed through the selective isolation of thrombin from fetal bovine serum and bovine plasma spiked with the thrombin standard, as confirmed by sodium dodecyl sulfate-polyacrylamide (SDS-PAGE) analysis.



INTRODUCTION

Thrombin (TB), a significant serine proteolytic enzyme, plays a pivotal role in a series of biological events.[1,2] Clinically, TB is known as an important blood clotting factor, which facilitates the conversion of soluble fibrinogen into insoluble fibrin in plasma.[3] In addition, the level of TB in humans is associated with the occurrence or development of some diseases, such as, thrombosis, inflammation, atherosclerosis, and myocardial infarction.[4,5] Consequently, thorough analysis of TB is highly important for the early diagnosis and treatment of related diseases and its clinical application. However, TB belongs to low-abundance proteins, and the concentration level of TB in plasma is quite limited.[6,7] Meanwhile, the highly complex sample matrix also presents great challenges for TB analysis. Consequently, developing efficient methodologies to selectively separate and enrich TB from real biological samples is of great importance.

Solid-phase extraction (SPE) is an effective approach for separating the desired analyte from a complex matrix.[8] In comparison with other conventional techniques for protein peptide separation, e.g., isoelectric precipitation, organic solvent precipitation, ultrafiltration, and electrophoresis, etc., SPE shows outstanding merits, including simple manipulation, high separation efficiency, and remarkable selectivity.[9,10] Various tailor-designed materials have exhibited favorable performance in the selective isolation of target protein species.[11−14]

Recently, dendritic mesoporous silica nanoparticles (DMSNs) with a three-dimensional dendritic framework structure have attracted extensive interest.[15] As a novel type of porous silica material, their remarkable characteristics of open central-radial pore structure, highly accessible surfaces, tunable size and morphology, ease of functionalization, and good biocompatibility[16−18] render them widely applicable in various fields, such as adsorption, catalysis, and drug delivery.[19−22] The unique central-radial pore structure facilitates the diffusion of biomacromolecules into the pores, allowing both the internal and external surfaces of the material to interact effectively with the target biomacromolecules. Therefore, dendritic mesoporous silica-based materials hold promising potential for the adsorption and separation of biomacromolecules such as proteins and peptides.[23−25]

Heparin is a kind of highly sulfated linear polysaccharide consisting of glucosamine and either D-glucuronic acid or L-iduronic acid.[26,27] Heparin possesses a high density of negative charges due to the existence of carboxylic and sulfonate groups.[28] There are two cationic patches on the surface of TB (denoted as anion binding exosites I and II). The anion-binding exosite II is on the northwestern face of TB, and it is required for interaction with heparin.[29,30] Owing to the above properties,

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heparin offers the possibility for selective adsorption/isolation of TB.

Herein, novel heparinized dendritic mesoporous silica nanoparticles (DMSNs-NH-Hep) are prepared via grafting APTES and heparin into the dendritic structure. The large surface area and the open central-radial pore structure provide a lot of accessible affinity sites to interact with TB. Benefiting from the strong interaction between heparin and TB, DMSNs-NH-Hep exhibit excellent selectivity and favorable adsorption efficiency toward TB. The promising practical applications of DMSNs-NH-Hep are verified by the isolation of TB from TB-spiked fetal bovine serum and bovine plasma.

EXPERIMENTAL SECTION

Chemicals and Materials. Sodium dodecyl sulfate (SDS), 4-morpholineethanesulfonic acid hydrate (MES), 1-(3-(dimethylamino)propyl)-3-ethylcarbodiimide hydrochloride (EDC), tetraethyl orthosilicate (TEOS), N-hydroxysuccinimide (NHS), ammonium hydroxide ($NH_3{\cdot}H_2O$), sodium salicylate, hexadecyl trimethylammonium bromide (CTAB), heparin sodium (Hep), bovine serum albumin (BSA), tris-(hydroxymethyl)aminomethane (Tris), and Coomassie brilliant blue G-250 are obtained from Macklin Biochemical Co., Ltd. (Shanghai, China). Glycine, thrombin (TB) from bovine plasma, and an SDS-PAGE gel quick preparation kit are acquired from Dalian Meilun Biotechnology Co., Ltd. (Dalian, China). The protein molecular weight marker (broad, D532A; Takara Biotechnology Company, Dalian, China) is a mixture of nine purified proteins (Mw in kDa: myosin, 200; b-galactosidase, 116; phosphorylase B, 97.2; serum albumin, 66.4; ovalbumin, 44.3; carbonic anhydrase, 29; trypsin inhibitor, 20.1; lysozyme, 14.3; aprotinin, 6.5). Triethanolamine (TEA), trypsin (Try), and (3-aminopropyl)triethoxysilane (APTES) are supplied by Aladdin Chemical Reagent Co., Ltd. (Shanghai, China). Ethanol, methanol, sodium chloride (NaCl), sodium hydrogen phosphate dodecahydrate ($NaHPO_4{\cdot}12H_2O$), potassium dihydrogen phosphate (KH_2PO_4), potassium chloride (KCl), and glacial acetic acid (HAc) are obtained from Damao Chemicals (Tianjin, China). Transferrin (Trf) is purchased from Sigma-Aldrich (St Louis, MO, USA). Hemoglobin (Hb) from bovine blood is acquired from J&K Scientific Ltd. (Beijing, China). Deionized water is used throughout the experiments.

Instruments. The morphology and size information of the nanoparticles are examined using a Sigma 360 scanning electron microscope (SEM, Zeiss, Germany) equipped with an energy-dispersive X-ray (EDX) spectroscopy system. The surface charge properties and dynamic light scattering (DLS) particle size distributions of the nanoparticles are analyzed using a Zetasizer Nano ZS90 (Malvern, UK). Zeta potential measurements are performed at pH 7 and room temperature. The solid-state ^{13}C nuclear magnetic resonance (NMR) spectrum of DMSNs-NH-Hep is obtained on an AVANCE III HD 400 M spectrometer (Bruker, Germany). Transmission electron microscopy (TEM) images of the materials are observed by a JEM-F200 transmission electron microscope (JEOL, Japan). X-ray photoelectron spectroscopy (XPS) spectra of DMSNs-NH_2 and DMSNs-NH-Hep are acquired using a K-Alpha X-ray photoelectron spectrometer (Thermo Scientific, USA). Nitrogen adsorption/desorption isotherms are obtained on an ASAP 2460 automatic gas adsorption analyzer (Micromeritics, USA). The specific surface area of the materials is calculated by the Brunauer–Emmett–Teller (BET) method, and the pore size distribution curves are derived by using the Barrett–Joyner–Halenda (BJH) method.

Synthesis of DMSNs and DMSNs-NH_2. The dendritic mesoporous silica nanoparticles (DMSNs) are prepared by an anion-assisted synthesis method.[16,31] In general, 68 mg of triethanolamine is dissolved in 25 mL of deionized water, and the solution is stirred for 30 min at 80 °C. 168 mg of sodium salicylate and 380 mg of CTAB are added to the triethanolamine solution, and the reaction mixture is magnetically stirred at 80 °C for 1 h. Then, 4 mL of tetraethyl orthosilicate (TEOS) is slowly added to the mixture under continuous stirring. After further stirring for 2 h at 80 °C, the system is cooled to room temperature. The white solid product is collected by centrifugation and washed twice with anhydrous ethanol. To remove the surfactant templates (sodium salicylate and CTAB) from the material, the white product is dispersed into 50 mL of anhydrous methanol in a 100 mL round-bottom flask. Subsequently, 3 mL of hydrochloric acid is carefully added to the DMSNs suspension, and the mixture is refluxed at 60 °C for 6 h. The reflux process is repeated three times. After centrifugation, the collected product is washed with anhydrous ethanol several times and dried at 60 °C for further use.

100 mg of DMSNs is added to a beaker containing 50 mL of anhydrous ethanol, and a uniform suspension is obtained by ultrasonic treatment. 800 μL of $NH_3{\cdot}H_2O$ and 800 μL of APTES are added dropwise to the above suspension under stirring. Then, the mixture is stirred at room temperature for 11 h. After centrifugation, the obtained product is washed with ethanol and dried under vacuum at 50 °C to obtain amino-terminated DMSNs (DMSNs-NH_2).

Synthesis of DMSNs-NH-Hep. 60 mg of heparin sodium, 60 mg of EDC, and 60 mg of NHS are first dissolved in 10 mL of MES buffer solution (10 mM, pH 6.0). After magnetic stirring for 30 min at room temperature, the pH of the solution is adjusted to 7.0 using a 5 mol/L NaOH solution. 30 mg of DMSNs-NH_2 powder is dispersed in 0.5 mL of water in advance, and the DMSNs-NH_2 suspension is mixed with the above solution. The mixture is stirred at room temperature for 4 h. The product is collected by centrifugation and washed several times with deionized water. Finally, the heparinized dendritic mesoporous silica nanoparticles (DMSNs-NH-Hep) are obtained after freeze-drying.

Adsorption/Desorption of Proteins by DMSNs-NH-Hep. Thrombin, trypsin (Try), and three other major proteins in blood—bovine serum albumin (BSA), transferrin (Trf), and hemoglobin (Hb)—are used as model proteins to study the adsorption performance of DMSNs-NH-Hep. A 4 mg/mL suspension of DMSNs-NH-Hep is prepared by dispersing the DMSNs-NH-Hep in PBS buffer solution (pH 7.4, 10 mM) by ultrasonication. 15 μL of the suspension (0.06 mg) is transferred into a 0.5 mL centrifuge tube and mixed with 85 μL of protein solution (dissolved in PBS buffer at pH 7.4). The mixture is shaken at room temperature for 3 min to promote the adsorption of proteins onto the material. The supernatant is collected after centrifugation at 8000 rpm for 5 min and mixed with Coomassie Brilliant Blue G250. The absorbance of the stained solution is measured at 595 nm to quantify the residual protein in the solution and calculate the corresponding adsorption efficiency.

For the recovery of adsorbed proteins from DMSNs-NH-Hep, 0.3% SDS solution is adopted as the eluent. DMSNs-NH-Hep, collected after the adsorption process, is prewashed with 100 μL of deionized water. Then, 100 μL of a 0.3% SDS solution

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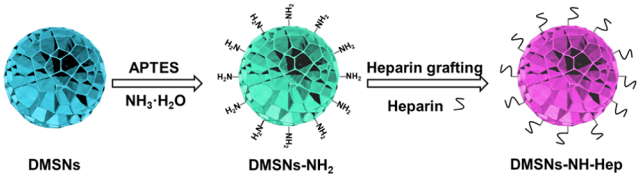
is added, and the mixture is vigorously shaken for 30 min to elute the proteins adsorbed on the material. After the elution, the supernatant is collected for calculating the elution efficiency or conducting subsequent research. The protein content in the supernatant is measured using the bicinchoninic acid (BCA) assay by determining the absorbance at 562 nm.

Isolation of TB from Complex Samples by DMSNs-NH-Hep. Fetal bovine serum (FBS) and bovine plasma spiked with the TB standard are adopted to investigate the adsorption performance of DMSNs-NH-Hep toward TB under complex biological conditions. The serum sample is diluted with PBS buffer solution (pH 7.4, 10 mM). Specifically, 0.08 mg of DMSNs-NH-Hep is mixed with 100 μL of 100-fold diluted fetal bovine serum (containing 10 U of TB), and the mixture is shaken for 3 min to facilitate the adsorption of TB. After the adsorption process, DMSNs-NH-Hep are collected by centrifugation (8000 rpm, 5 min) and washed three times with 100 μL of deionized water. Subsequently, 100 μL of 0.3% SDS solution is used to elute the TB captured by DMSNs-NH-Hep. The protein solutions from each step are collected and subjected to SDS-PAGE analysis. When bovine plasma is used, the experimental procedure is the same as above, except that FBS is replaced with bovine plasma.

RESULTS AND DISCUSSION

Characterization of DMSNs-NH-Hep. The synthetic strategy for DMSNs-NH-Hep, based on the surface grafting approach, is illustrated in Scheme 1. An anion-assisted method is adopted for the synthesis of dendritic mesoporous silica nanoparticles (DMSNs), with TEOS as precursors and CTAB and sodium salicylate as structure-directing agents. Then, the silanization reagent APTES is adopted to introduce amino groups into the structure of DMSNs, and the resulting amino-terminated DMSNs are named DMSNs-NH_2. Finally, heparinized dendritic mesoporous silica nanoparticles (DMSNs-NH-Hep) are synthesized via carbodiimide chemistry.[32]

Scheme 1. Schematic Illustration for the Preparation of DMSNs-NH-Hep



The surface morphology and structure of DMSNs-NH_2 and DMSNs-NH-Hep were observed using scanning electron microscopy (SEM) and transmission electron microscopy (TEM). As shown in the SEM image of DMSNs-NH_2 (Figure 1A), DMSNs-NH_2 exhibit a spherical shape and a uniform size distribution. Numerous irregularly shaped nanopores are distributed on the surface of DMSNs-NH_2 (Figure 1A). The unique dendritic framework structure with a central-radial pore structure is clearly demonstrated in the TEM image of DMSNs-NH_2 (Figure 1C). As shown in Figure 1B, DMSNs-NH-Hep nearly retains the morphology of the initial DMSNs-NH_2 without noticeable change. The TEM image of DMSNs-NH-Hep (Figure 1D) illustrates that the open pore structure is not disrupted during the modification process. Dynamic light scattering (DLS) was used to examine the particle size distributions of DMSNs, DMSNs-NH_2, and DMSNs-NH-Hep. The DLS particle size distribution results are illustrated in Figure S1. The averaged diameters of DMSNs, DMSNs-NH_2, and DMSNs-NH-Hep are 275.0, 275.2, and 268.3 nm, respectively. The modification with APTES and heparin did not lead to significant changes in particle size. The DLS particle size distribution results of DMSNs-NH_2 and DMSNs-NH-Hep are in agreement with those derived from SEM and TEM images (DMSNs-NH_2, SEM: 265.6 nm and TEM: 267.5 nm; DMSNs-NH-Hep, SEM: 266.5 nm and TEM: 269.0 nm).

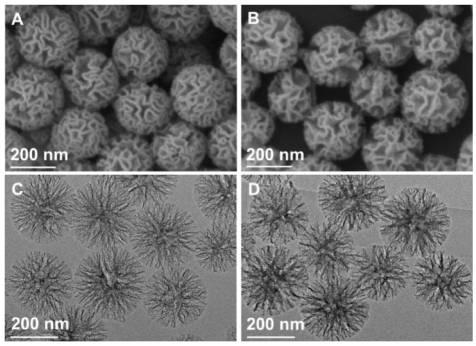


Figure 1. SEM images of DMSNs-NH_2 (A) and DMSNs-NH-Hep (B); TEM images of DMSNs-NH_2 (C) and DMSNs-NH-Hep (D).

The energy-dispersive X-ray spectroscopy (EDX) mapping results of DMSNs-NH-Hep (Figure 2) show that N, O, Si, and S are distributed within the DMSN-NH-Hep structure. Heparin is a polysaccharide containing abundant sulfonate groups, and the presence of S confirms the successful preparation of DMSNs-NH-Hep.

X-ray photoelectron spectroscopy (XPS) is applied to analyze the elemental composition of DMSNs-NH_2 and DMSNs-NH-Hep. As shown in the XPS spectrum of DMSNs-NH_2 (Figure 3A), peaks of O 1s, N 1s, C 1s, Si 2s, and Si 2p are observed, indicating the existence of O, N, C, and Si in DMSNs-NH_2. The presence of N indicates amino groups on the surface of DMSNs-NH_2. There is almost no difference in elemental compositions between DMSNs-NH_2 and DMSNs-NH-Hep, except for the existence of S in DMSNs-NH-Hep (Figure 3A). The S 2p peak at approximately 168.8 eV (Figure 3B) is ascribed to the sulfonate groups in the heparin structure.[33] In the high-resolution C 1s spectrum of DMSNs-NH_2 (Figure S2A), the peaks at 284.8 and 286 eV are assigned to C−C and C−N.[34] For DMSNs-NH-Hep, two new peaks ascribed to C−O (286.5 eV) and C═O (288 eV)[34] derived from heparin are observed (Figure S2B), demonstrating the existence of heparin in the nanoparticles.

Zeta potential analysis results in Figure 3C show that DMSNs are negatively charged. After amino modification by APTES, DMSNs-NH_2 exhibit a positive charge, owing to the introduction of amino groups. Heparin contains a high density of negative charge due to the existence of carboxylic and sulfonate groups.[28] Therefore, the zeta potential of DMSNs-NH-Hep becomes negative after the grafting of heparin.

The ^{13}C NMR spectrum of DMSNs-NH-Hep is illustrated in Figure 3D. The signals at 173.7 and 99.4 ppm are associated with the carbonyl group and the anomeric carbon in heparin, respectively. The resonance peaks at 69.9 and 57.4 ppm are assigned to the glucosamine residues of heparin.[35] The signals due to sp^3 carbons of aminopropyl groups are observed at 42.5, 21.3, and 9.5 ppm.[36] The information provided by the NMR

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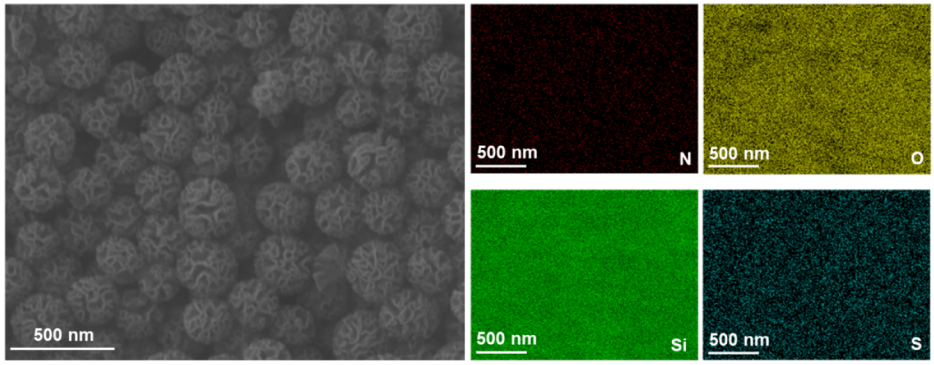


Figure 2. EDX elemental mapping results of DMSNs-NH-Hep.

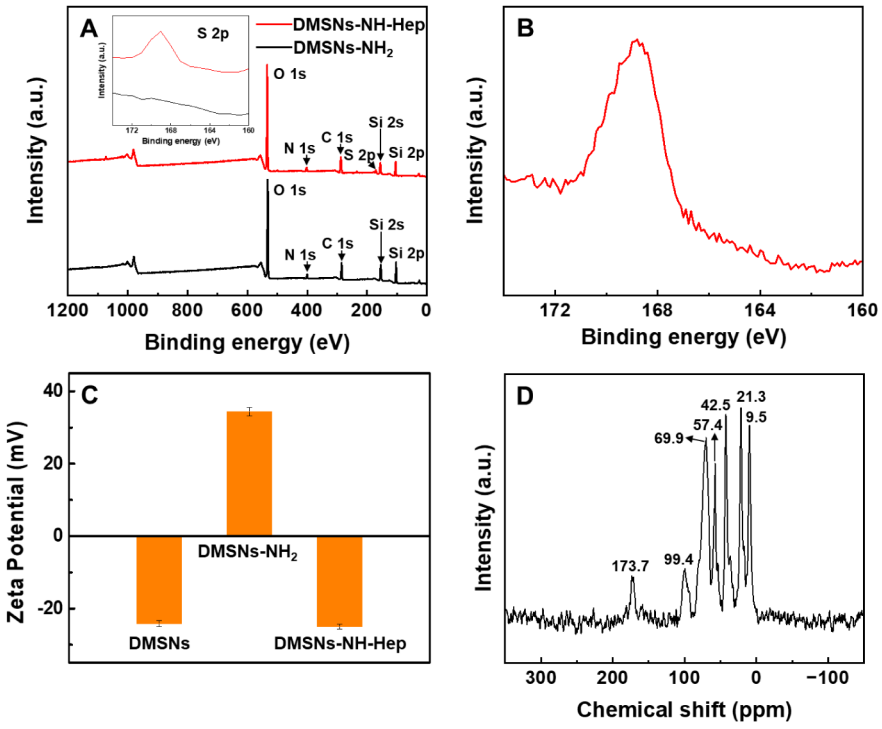


Figure 3. XPS spectra of DMSNs-NH_2 and DMSNs-NH-Hep, and the inset shows partial magnification spectra of S 2p (A); high-resolution S 2p spectrum of DMSNs-NH-Hep (B); zeta potential of DMSNs, DMSNs-NH_2, and DMSNs-NH-Hep (C); ^{13}C NMR of DMSNs-NH-Hep (D).

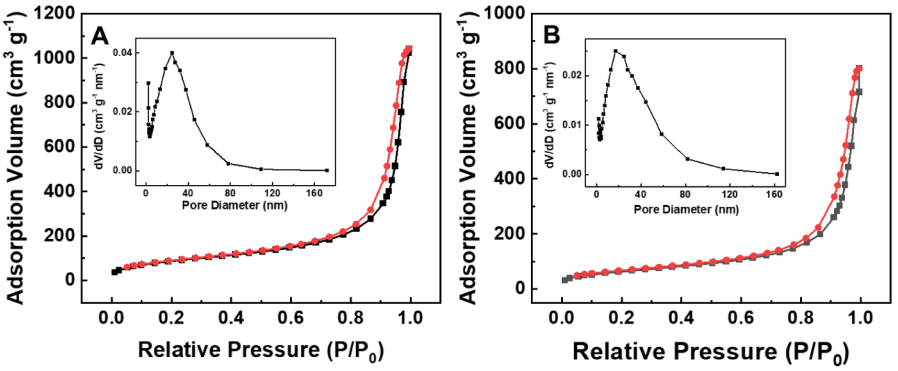


Figure 4. Nitrogen adsorption/desorption isotherms and BJH pore-size distribution curves for DMSNs-NH_2 (A) and DMSNs-NH-Hep (B).

spectra confirms the successful preparation of DMSNs-NH-Hep.

Nitrogen adsorption/desorption isotherms are conducted to study the textural properties of DMSNs-NH_2 and DMSNs-NH-Hep (Figure 4). The Brunauer−Emmett−Teller (BET) surface area and pore volume of DMSNs-NH_2 are 328.75 $m^2\ g^{-1}$ and 1.61 $cm^3\ g^{-1}$, with the pore size mainly centered at 24.6 nm. After the functionalization process, the BET surface area and

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pore volume of DMSNs-NH-Hep decrease to 230.81 $m^2 g^{-1}$ and 1.23 $cm^3 g^{-1}$, and the pores are mainly centered at 17.1 nm. The above results suggest that the obtained DMSNs-NH-Hep possess large pore size and high surface area. The large pore size of DMSNs-NH-Hep allows protein molecules to diffuse into the pore channels, providing abundant accessible affinity sites to interact with protein molecules.

Protein Adsorption Behavior by DMSNs-NH-Hep. Thrombin is a serine protease in the bloodstream, and albumin, transferrin, and hemoglobin are three major high-abundance proteins in blood, which are potential interfering factors for the isolation of TB. Trypsin (Try) is also a serine protease, and its structure is remarkably similar to thrombin.[37] To study the selectivity and purification ability of DMSNs-NH-Hep for TB, we investigated these model proteins. The adsorption performance of DMSNs-NH_2 and DMSNs-NH-Hep for these proteins was evaluated at pH 7.4. As shown in Figure 5, DMSNs-NH-Hep exhibited an adsorption efficiency of 81.1% for TB, while DMSNs-NH-Hep demonstrated little adsorption for BSA, Trf, Try, and Hb, with adsorption efficiencies of less than 30% is obtained. These results indicate that DMSNs-NH-Hep possess excellent adsorption selectivity and efficiency for TB. When DMSNs-NH_2 were used as adsorbents, the adsorption efficiency of DMSNs-NH_2 for TB was 15.3%, and a higher adsorption efficiency of 61.3% for BSA was obtained. It is obvious that the introduction of heparin in DMSNs-NH-Hep can improve the selective adsorption performance for TB. There are two anion-binding exosites in the TB structure, and the anion-binding exosite II contains Arg93, Arg101, Arg126, Arg165, Arg233, Lys235, Lys236, and Lys240 residues.[38] The anion-binding exosite II is located on the northwestern face of TB and is required for interaction with heparin.[29,30] The strong interaction between heparin and the anion binding exosite II promotes the selective adsorption of TB by DMSNs-NH-Hep.

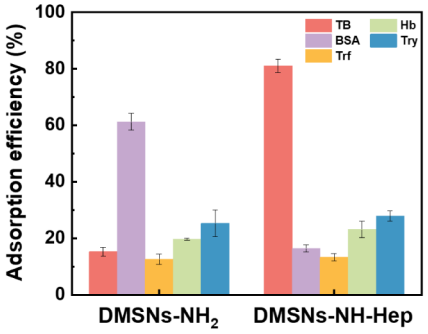


Figure 5. A comparison of DMSNs-NH_2 and DMSNs-NH-Hep for the adsorption of TB, BSA, Trf, Try, and Hb. Protein solution: TB: 0.1 mL, 85 U/mL; other proteins: 0.1 mL, 85 μg/mL; mass of adsorbent: 0.06 mg; adsorption time: 3 min; error bars indicate standard deviations (n = 3).

The adsorption time is a significant influencing factor for TB adsorption. The influence of the contact time on the adsorption of TB is illustrated in Figure 6A. The adsorption efficiency of DMSNs-NH-Hep for TB can reach 80.8% with an adsorption time of 1 min, indicating that the adsorption of TB by DMSNs-NH-Hep occurs quickly. In the subsequent adsorption experiments, an adsorption time of 3 min is adopted.

The adsorption capacity of DMSNs-NH-Hep for TB was evaluated using 0.015 mg of DMSNs-NH-Hep with a series of TB concentrations (17−68 U mL^{-1}) at room temperature. The amount of adsorbed TB (Q, U mg^{-1}) is evaluated by the following equation, where V (mL), C_0 (U mL^{-1}), C_1 (U mL^{-1}), and M (mg) represent the volume of the sample solution, the initial protein concentration, the protein concentration after the adsorption process, and the mass of DMSNs-NH-Hep.

$$Q = V(C_0 - C_1)/M$$

As demonstrated in Figure 6B, DMSNs-NH-Hep provide an adsorption capacity of 237.35 U mg^{-1} in 3 min for TB. Table 1 summarizes the adsorption capacities of reported materials for TB. The prepared DMSNs-NH-Hep exhibits remarkable adsorption performance for TB with respect to other reported adsorbents. This can be attributed to the unique porous structure and highly accessible surface of DMSNs-NH-Hep,

Table 1. Comparison of Maximum Adsorption Capacity for TB with Diverse Materials

Materials	Adsorption capacity (U mg^{-1})	Ref.
Polypyrrole membrane	5.935	39
Heparin-coupled affinity electromembrane	10.301	40
Fe_3O_4 microsphere	6.2	41
Three-dimensional ordered macroporous composite	40.9	42
DMSNs-NH-Hep	237.35	This work

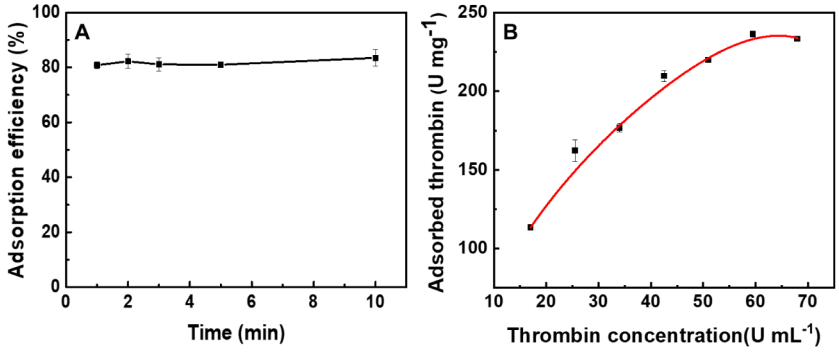


Figure 6. (A) The dependence of TB adsorption behavior on the adsorption time. Protein solution: 0.1 mL, 85 U/mL; mass of adsorbent: 0.06 mg. (B) The adsorption isotherm of DMSNs-NH-Hep for TB. Protein solution: 0.1 mL, 17−68 U/mL; mass of adsorbent: 0.015 mg; adsorption time: 3 min. Error bars indicate standard deviations (n = 3).

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which provide numerous accessible binding sites for TB adsorption.

Recovery of the Adsorbed TB from DMSNs-NH-Hep. For practical biological research, it is of great significance to recover adsorbed TB from DMSNs-NH-Hep into an aqueous medium. Several potential stripping reagents, e.g., 0.1% $NH_3 \cdot H_2O$, 0.5% $NH_3 \cdot H_2O$, 8 mol/L urea, 0.1 mol/L Tris, and 0.3% SDS, are employed for the stripping of TB. The desorption experiment results in Figure 7 reveal that 0.3% SDS demonstrates the best elution effect, with an elution efficiency of 80.4%. Therefore, 0.3% SDS is selected as the elution agent for subsequent studies.

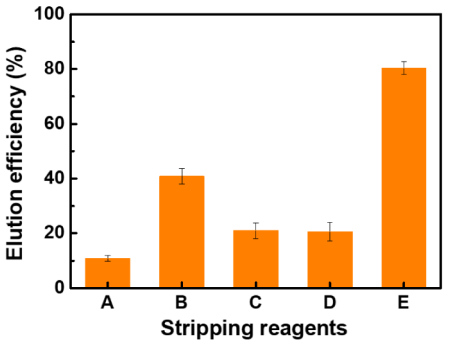


Figure 7. Elution efficiencies for the captured TB from DMSNs-NH-Hep with different stripping reagents: A: 0.1% $NH_3 \cdot H_2O$; B: 0.5% $NH_3 \cdot H_2O$; C: 8 mol/L urea; D: 0.1 mol/L Tris; E: 0.3% SDS. Protein solution: 0.1 mL, 85 U/mL; mass of adsorbent: 0.06 mg; adsorption time: 3 min; volume of stripping reagent: 0.1 mL; stripping time: 30 min; error bars indicate standard deviations (n = 3).

Isolation of TB from Complex Samples by DMSNs-NH-Hep. To demonstrate the separation and purification capability of DMSNs-NH-Hep for TB under complex biological conditions, DMSNs-NH-Hep was used for the selective adsorption of TB from TB-spiked fetal bovine serum. Figure 8

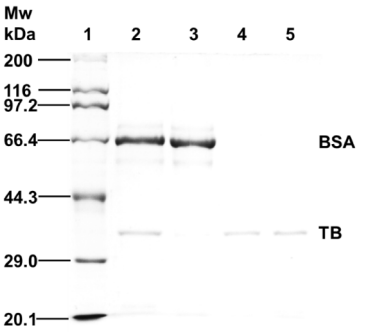


Figure 8. SDS-PAGE assay results: Lane 1: molecular weight marker (kDa); Lane 2: fetal bovine serum (100-fold dilution) added with TB standard solution; Lane 3: the supernatant of fetal bovine serum (100-fold dilution) added with TB standard solution after treatment with DMSNs-NH-Hep; Lane 4: TB recovered from DMSNs-NH-Hep; Lane 5: TB standard solution.

shows the SDS-PAGE results: in the initial TB-spiked FBS sample (lane 2), the protein bands mainly corresponding to BSA and TB are observed. After stripping the adsorbed proteins from DMSNs-NH-Hep with 0.3% SDS, only the band of TB can be observed in lane 4, and its position aligns with the standard TB band in lane 5. Bovine plasma was adopted to further test the selectivity of DMSNs-NH-Hep in the isolation of TB under complex biological conditions (Figure S3). Only a single band corresponding to TB is observed in the supernatant after elution (lane 4). The SDS-PAGE results demonstrate that DMSNs-NH-Hep can selectively separate TB from complex biological conditions.

CONCLUSIONS

In this study, heparinized dendritic mesoporous silica nanoparticles are successfully synthesized and characterized by SEM, EDX, TEM, DLS, XPS, zeta potential, NMR, and BET. The unique pore structure of DMSNs-NH-Hep makes it feasible to provide numerous accessible affinity sites for TB binding, resulting in a superior adsorption capacity of 237.35 U mg^{-1} for TB. Attributed to the strong affinity between heparin and TB, the constructed DMSNs-NH-Hep exhibit favorable adsorption selectivity for TB. Moreover, the SDS-PAGE results demonstrate that the prepared nanoparticles can successfully isolate TB from complex biological samples with high purity. The attempt in this work extends the application of porous silica nanoparticles in biological isolation and demonstrates a new avenue for the exploitation of effective adsorbents with excellent selectivity for specific biomacromolecules.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsomega.5c07418.

DLS particle size distributions of DMSNs, DMSNs-NH_2, and DMSNs-NH-Hep; high-resolution XPS C 1s spectra of DMSNs-NH_2 and DMSNs-NH-Hep; and SDS-PAGE assay results of bovine plasma (PDF)

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Notes

The authors declare no competing financial interest.

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